Si Clash Competition Final Report

Submitted by team 4 members:

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In this project, it's required to design an active, fully differential filter using op-amp (or OTA) RC filter stage in CMOS. A Gm-C filter may not be used. The design must be closed loop.

Check Point 1

What filter order and family best meet the desired specifications (Butterworth, Chebychev, Inverse-Chebychev, Bessel, ...)?

After reviewing filter theory literature, it has been found that the Bessel response gives the best group delay variation, which is required to be less than 3nsec. On the other hand, however, its stopband characteristics are relatively not enough to satisfy the attenuation requirements. Therefore, the decision has been made to use a "heterogeneous" response, which gives a linear phase response with 0.5° error, satisfying both group delay variation and stopband attenuation requirements. It has been found, after inspecting and modeling several choices, that the order satisfying required specs is 3rd order. Filter design table and equations used can be found in 1, page 50 and page 92, respectively.

Let

$$C_5 = 1\,\text{pF}$$

$$k = 2\pi f_0 C_5 = 2\pi \times 1.3292 \times 20\text{M} \times 1\text{p} = 1.67 \times 10^{-4}$$

$$C_2 = 4Q^2(H+1) = 4 \times 0.9536^2(1+1) \times 1\text{p} = 7.27\,\text{pF}$$

$$R_1 = R_4 = \frac{1}{2Qk} = \frac{1}{4 \times 0.9536 \times 1.67 \times 10^{-4}} = 3139\,\Omega$$

$$R_3 = \frac{1}{4Qk} = \frac{1}{4 \times 0.9536 \times 1.67 \times 10^{-4}} = 1570\,\Omega$$

RC 1st order section

$$f_0 = f_n \times BW = 0.8257 \times 2\pi \times 20\text{M} = \frac{1}{RC}$$

For

$$C = 500\,\text{fF} \quad \Rightarrow \quad R = 19.2\,\text{k}\Omega$$

Selected filter topology component values

Component	Value
C_5	1 pF
C_2	7.27 pF
$R_1 = R_4$	3139 Ω
R_3	1570 Ω
R	19.2 kΩ
C	500 fF

One may wonder: this 1pF initial assumption that all further calculations are based on is relatively large, isn't it? After all, this capacitor will occupy area, which is an important resource that should be saved till the last possible droplet. Well, if the design equations are carefully investigated, it will be clear that if smaller capacitance value is initially assumed, then the resistances will be of significant values that will add noise, and area can be paid to get achieved requirements.

Choose the filter topology. You can use Multiple Feedback, Biquads, or Tow-Thomas biquads, or others.

The goal is to choose a topology having the smallest possible number of Op Amps, to avoid excessive power consumption, large area and high noise floor that may violate the dynamic range spec. Also, the search range was limited by a topology that can be implemented in fully differential form easily. There's no topology that satisfies all these requirements better than the multiple feedback one.

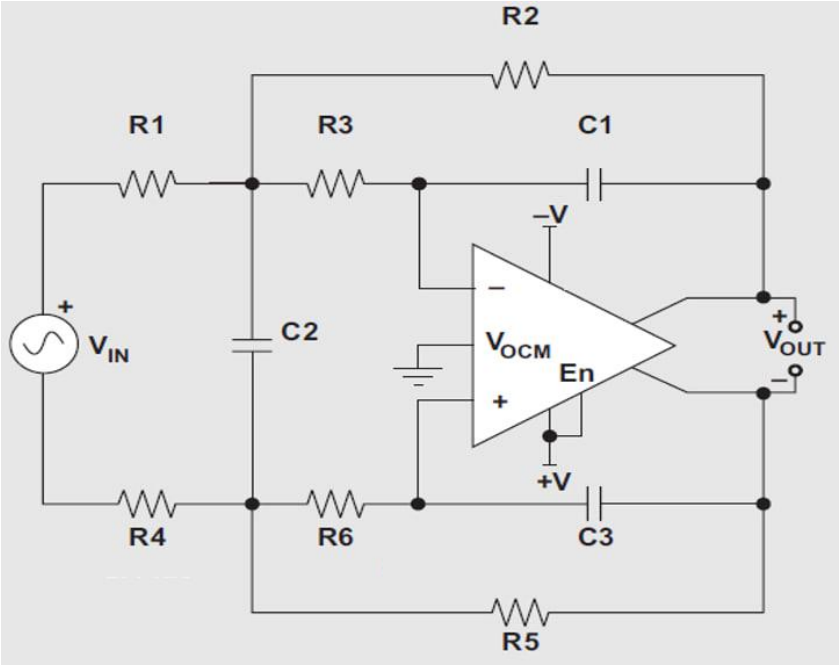


Figure 1: Fully differential multiple feedback (MFB) topology example.

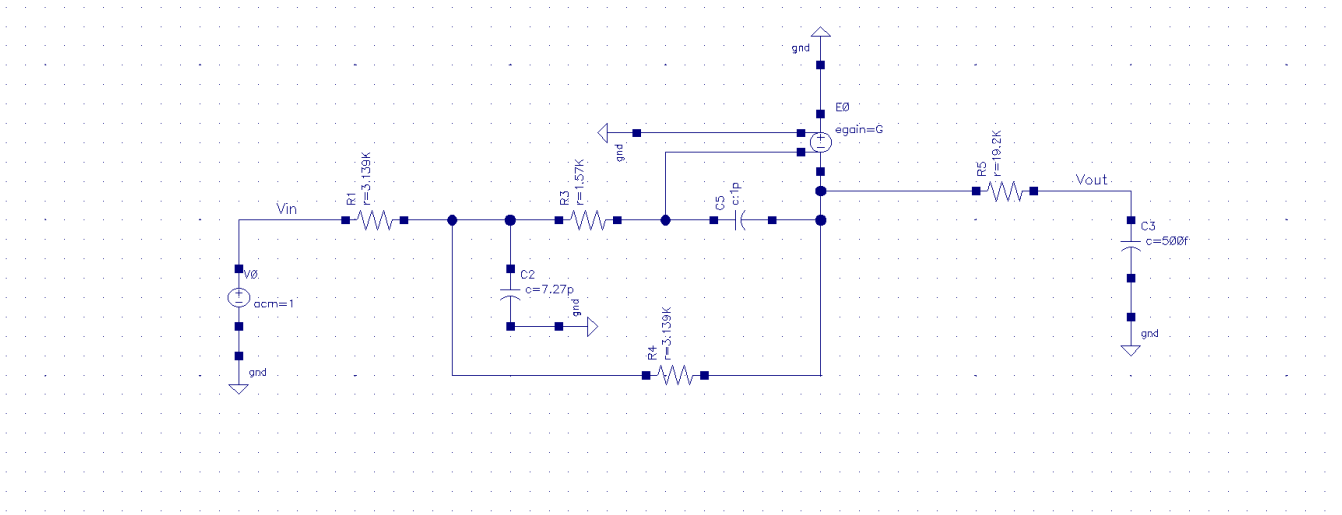


Figure 2: Behavioral testbench for the selected topology. Made SE just for simplicity.

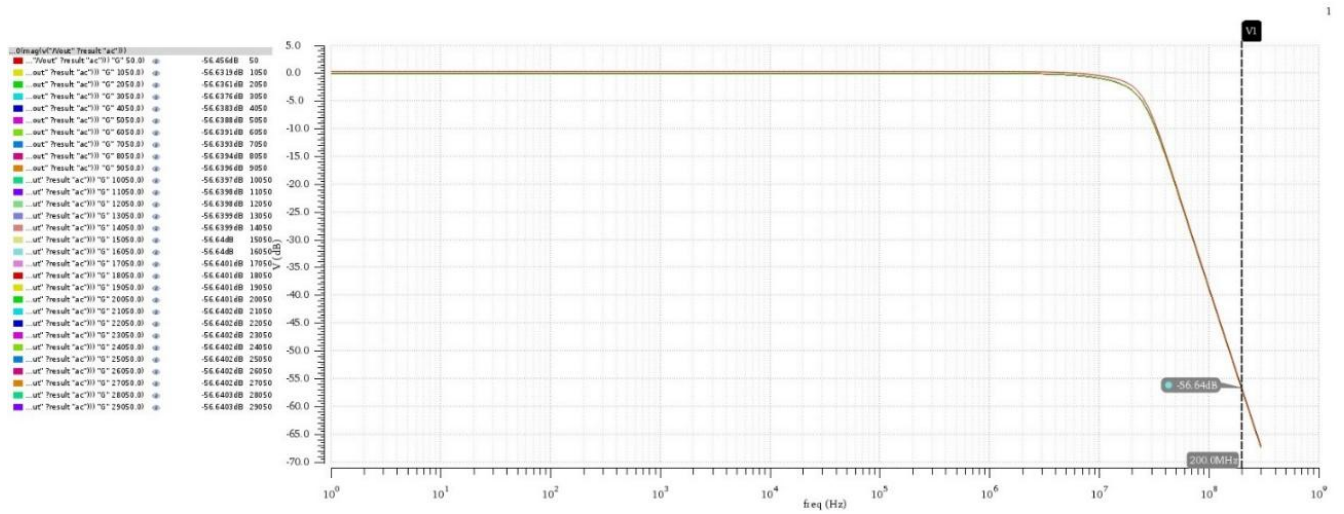


Figure 3: Required attenuation satisfied.

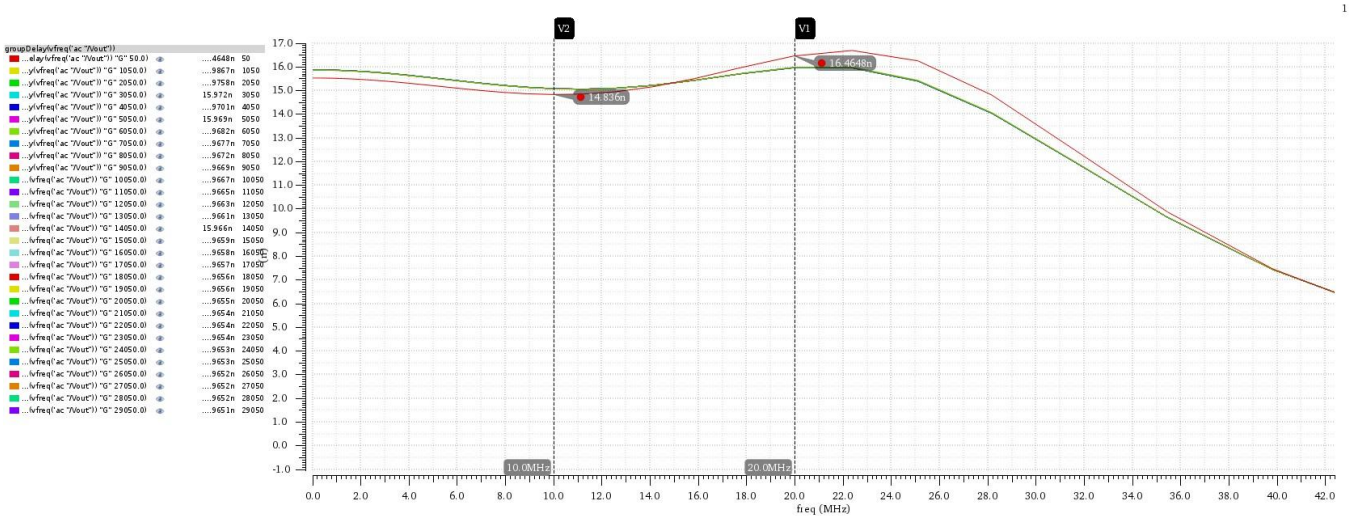


Figure 4: Max in band group delay ≈ 1.6nsec, annotated at the curve corresponding to gain of 50.

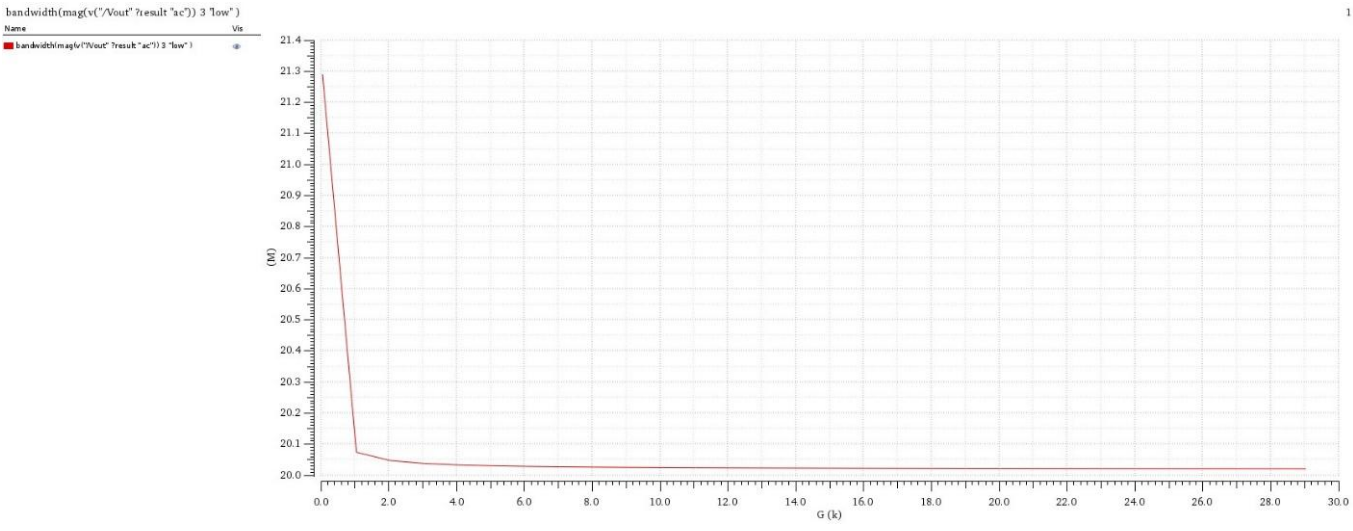


Figure 5: BW variation Vs gain.

Note that all curves are plotted Vs gain, parametrically swept from 50 to 30,000 with step of 1000.

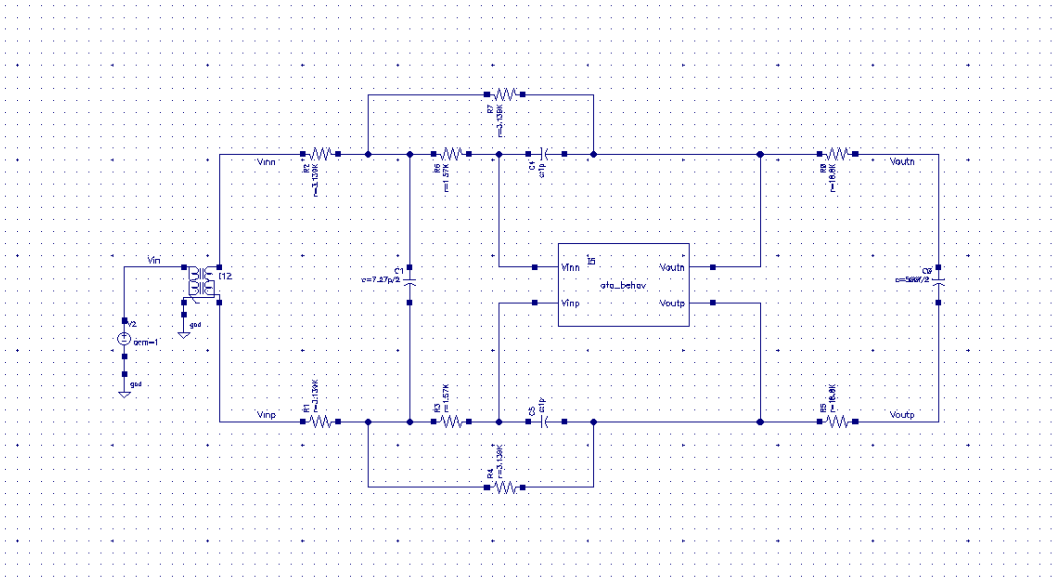


Figure 6: Modified testbench having a behavioral Op Amp at a lower level of abstraction.

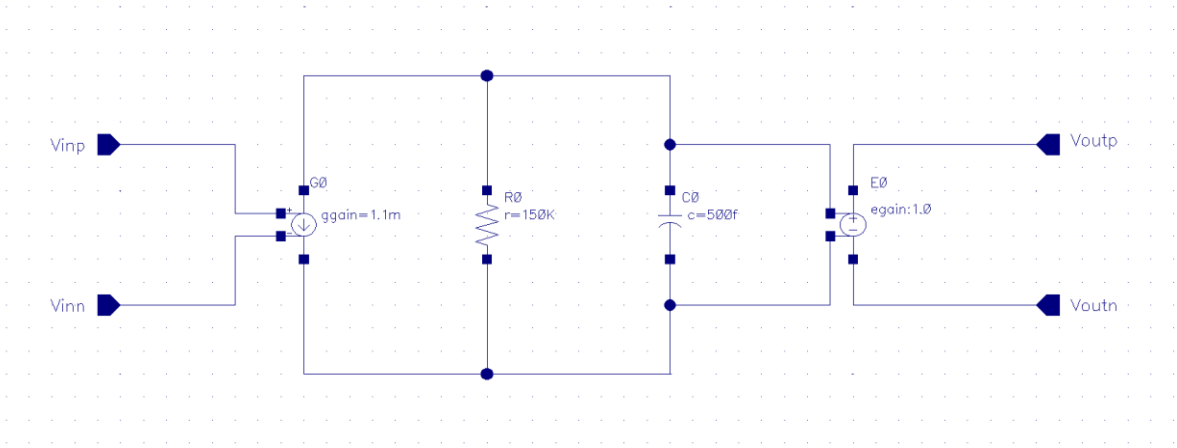


Figure 7: Behavioral model of the Op Amp.

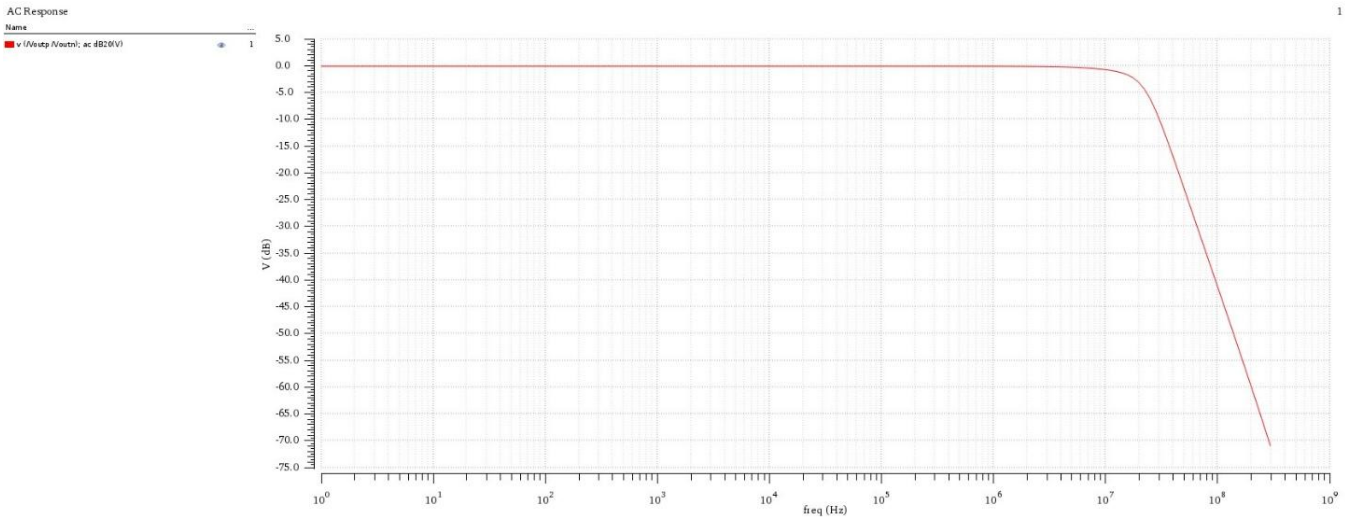


Figure 8: Amplitude frequency response.

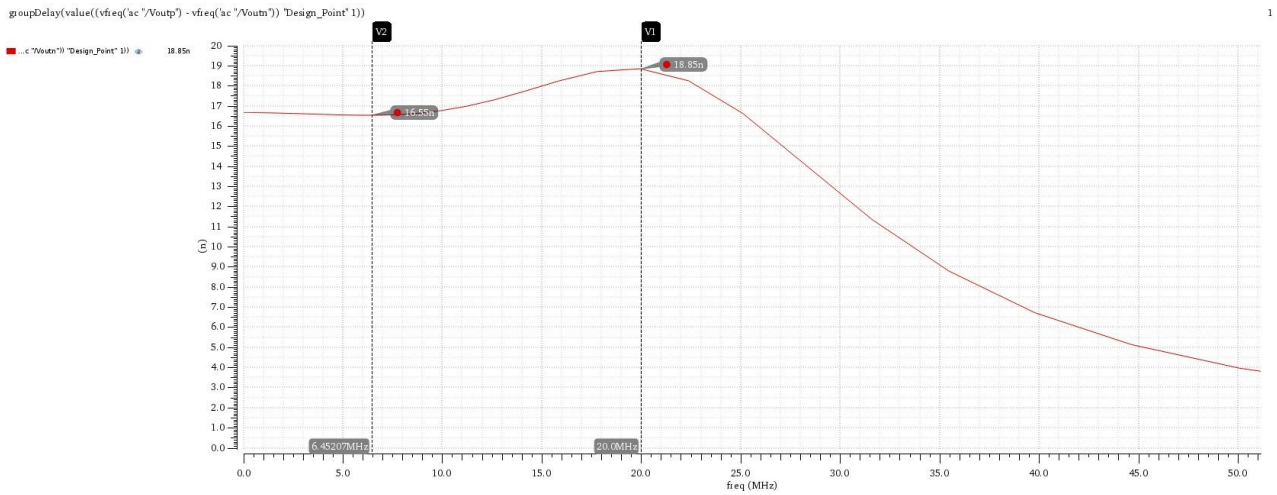


Figure 9: Max in band group delay ≈ 2.3nsec.

	Expression	Value
1	bandwidth(ma...	20.01E6

Figure 10: achieved BW.

A trial to implement the filter using Tow-Thomas biquads, which are much more immune to component variations than the MFB, was executed. However, unfortunately, it was found that the GBW required to satisfy specs sheet in the former is much more than that of the latter, corresponding to much more power consumption, so the decision was made to pay component variation immunity to buy less power dissipation. The details of this trial can be found in this file.

For the active stage, choose between an op-amp or an OTA. Explain your choice. You may choose to have a combination of OTA/op-amps for different stages.

Op Amp-based active stage is to be employed. Although consuming larger amounts of power when compared to OTAs, the buffer stage, which differentiates between Op Amps and OTAs, is necessary to avoid loading effect resulting from the resistors present in the feedback loop. The following figure illustrates one of the proposed buffer stages, the flipped voltage follower, which is characterized by a very low output resistance 2.

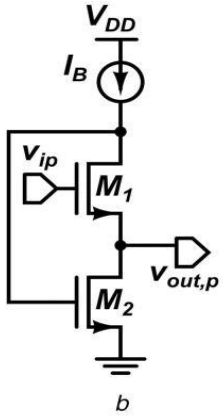


Figure 11: FVF output stage.

What are the OTA/op-amp specifications in terms of gain-bandwidth, gain, output resistance, and other specifications? Specify each stage.

$$GBW = 200\,\text{MHz} \quad \text{(as a rule of thumb)}$$

To obtain a reasonable estimation of parasitic capacitances that will be used in further analysis, a parametric sweep over the width and the length, and hence the area of the devices, was executed to extract different values of C_{dd} and the average was taken, the value obtained was as follows:

	Expression	Value
1	average(average(average(getData("M0:cdd" ?result "dc"))))	13.87E-15

	Expression	Value
1	average(average(average(getData("M1:cdd" ?result "dc"))))	12.45E-15

Figure 12: Average values of extracted parasitic capacitances.

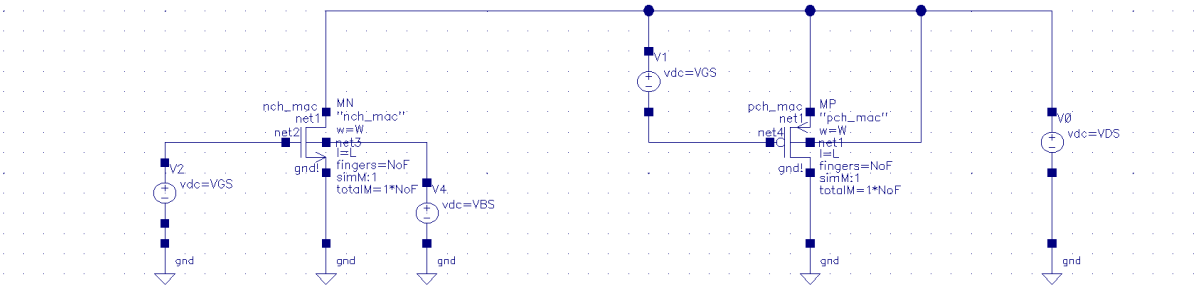


Figure 13: Testbench used to calculate parasitics.

Given:

$$GBW = 200\,\text{MHz} = \frac{G_{mi}}{2\pi C_{out}} = \frac{G_{mi}}{2\pi \times (40+60)\,\text{fF}} \quad \rightarrow \quad g_{mi} \approx 125\,\mu\text{S}$$

The parasitics are considered to be 60 fF to take some margin.

Another one spec of special importance is the output resistance. When sweeping over the behavioral model depicted in figure 6, it's been found that the highest output resistance that maintains the required 200MHz stopband attenuation and 3nsec group delay is 1kΩ.

When using $\frac{g_m}{I_D}$ (the notation TE "transistor efficiency" will be used rather than $\frac{g_m}{I_D}$ for convenience) design methodology, there are some well-known rules of thumbs that are followed in the design procedure. From gain and noise perspective, it is recommended to use large TE for devices whose g_m

contributes to voltage-to-current conversion, such as input and cascode transistors, while using smaller values of TE for other devices such as DC bias current sources.

Based on these rules of thumb, it is assumed that $TE_i = 15$.

Size resistors and specify input noise requirements for the OTA/op-amp to meet dynamic range requirements.

A trial to estimate the input-referred noise can also be found in this file.

Estimate the power consumption of your design based on these requirements. Do not run transistor level simulations on the entire circuit (except on a single device). Use general concepts to perform the design.

$$\text{Given } TE_i = 15,\ g_{mi} = 125\,\mu\text{S}:$$
$$\rightarrow \quad I_D \approx 8\,\mu\text{A} \quad \rightarrow \quad I_{SS} = 2I_D = 16\,\mu\text{A}$$

As a rough estimation for power consumption, it may be concluded that:

$$P_{dc} = V_{DD} \times I_{tot} = 1.2\,\text{V} \times 16\,\mu\text{A} \times 3 \approx 58\,\mu\text{W}$$

A factor of 3 is added to account for CMFB network and output stages that will be added.

Active Stage Transistor Level

1. Introduction

The choice was first to employ a telescopic cascode structure followed by a simple common source stage to form a two stage OTA along with an appropriate output stage to obtain an Op Amp, based on the assumption that the large GBW required can be readily achieved via using such high gain topology. When the initial testbenches results were investigated, however, it has been found that this topology is nothing but a waste of resources that, in addition, generates more noise (more devices) for nothing. It has been verified, while running simulations on these testbenches, that just a simple actively loaded differential pair provides roughly double the required GBW.

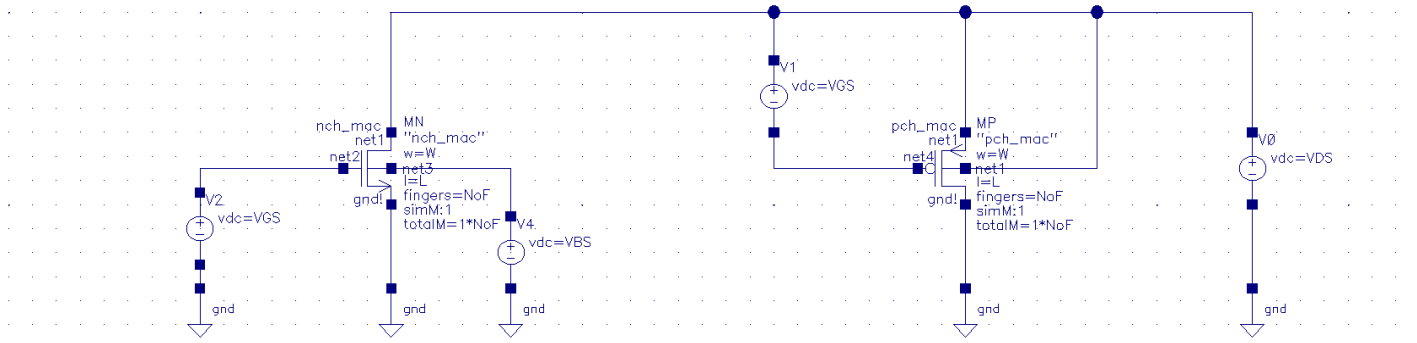


Figure 14: The testbench used to generate design charts.

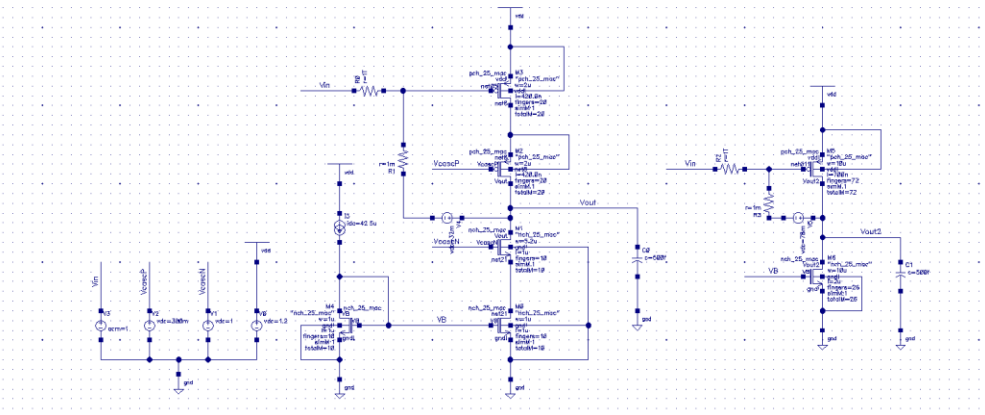


Figure 16: Initial testbench for two stage amp.

	Design_Point	gainBwP..."ac")))
1	1.000	163.8E6

	Design_Point	gainBwP..."ac")))
1	1.000	415.7E6

Figure 15: GBW obtained from cascode (left) and CS stage (right).

2. Core Amplifier Design

Following figures illustrate the core amplifier implementation and some of its key performance metrics.

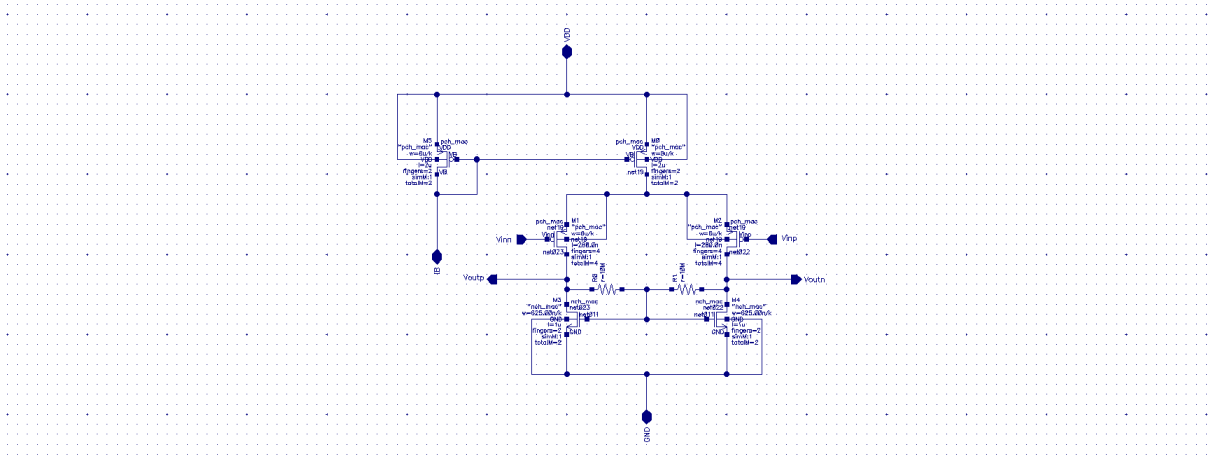


Figure 16: Schematic of core amplifier stabilized with behavioral CMFB network.

The input pair has been chosen to be of PMOS type since it has lower flicker noise that most probably is dominant in the band of operation, as shown in literature that the flicker noise corner can be as large as hundreds of MHz in deep submicron technologies. Paying attention to noise aspects of the design is of special importance since there's constraints on dynamic range. Resistors of 10MΩ resistance are used as a behavioral CMFB network.

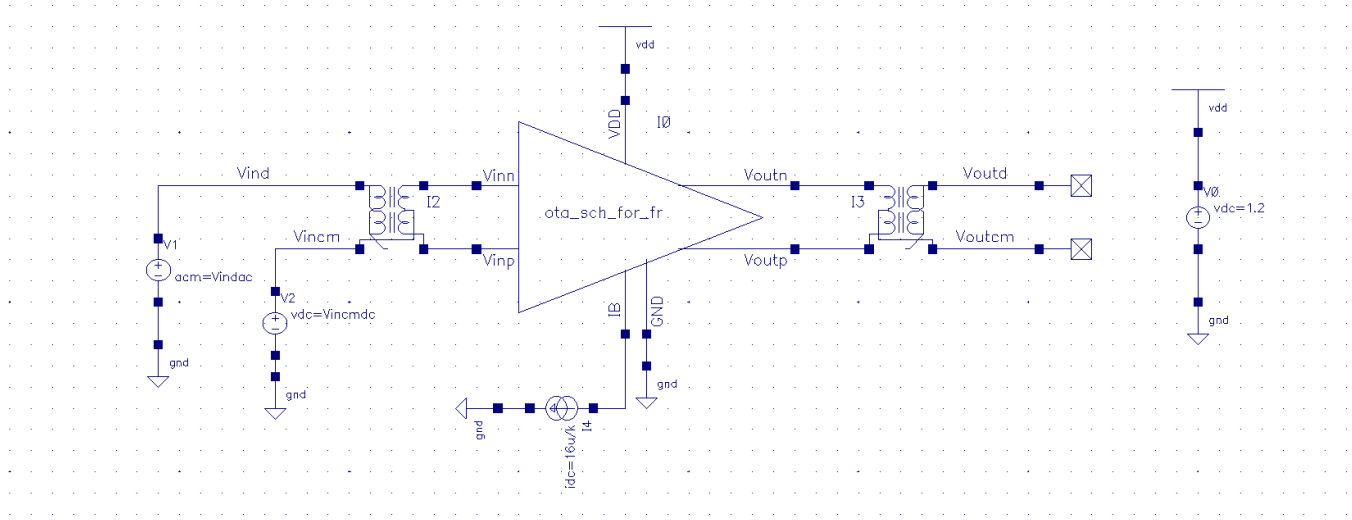


Figure 17: open loop testbench created to investigate core amp performance.

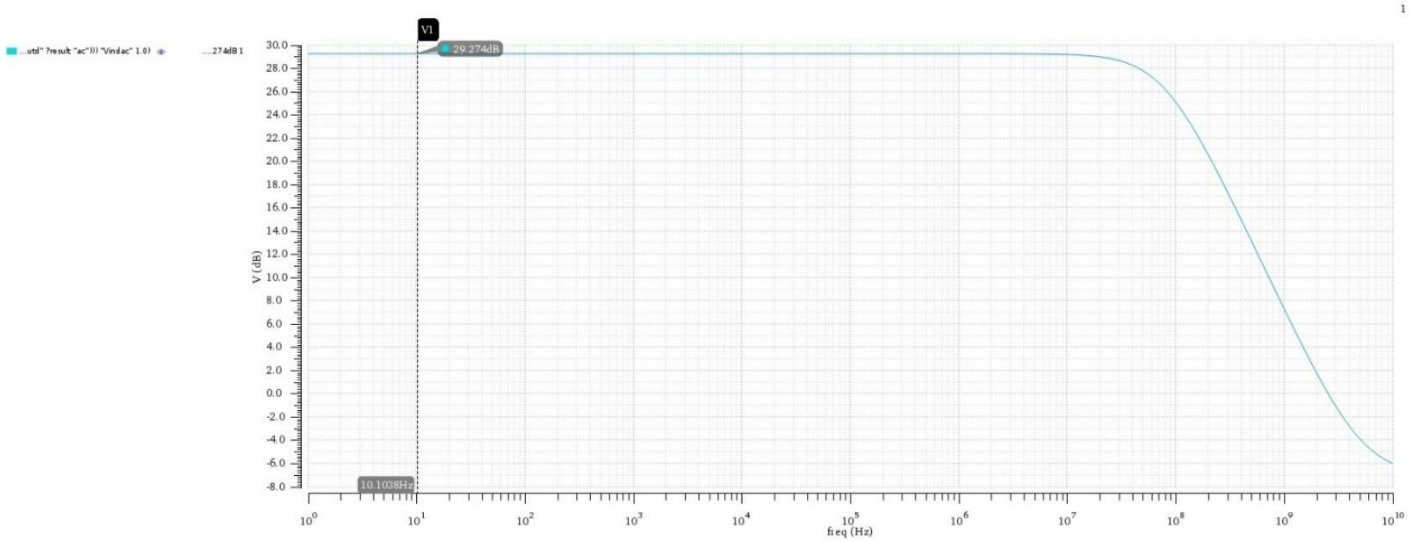


Figure 18: Amp DC gain.

	Vindac	gainBwP..."ac")))
1	1.000	2.432E9

	Vindac	V2:pwr (W)
1	1.000	-19.00E-6

Figure 19: achieved GBW with power consumption.

3. Auxiliary Circuit Design

3.1. Output Stage

Output stages, generally, seem to be easy to design and the required performance is extracted from it with the minimum effort. Actually, it has appeared that this is not the case. A multitude of source follower topologies were examined such as super source follower (SSF), flipped voltage follower (FVF), simple complementary source follower and resistive feedback inverter amplifier.

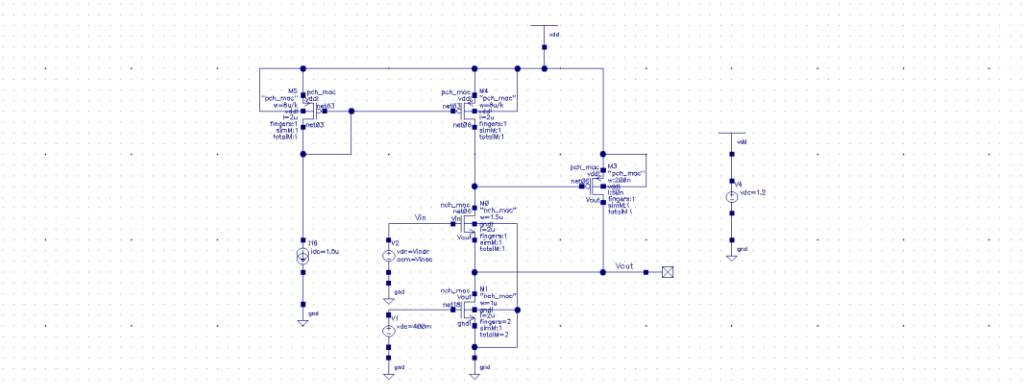


Figure 20: SSF testbench.

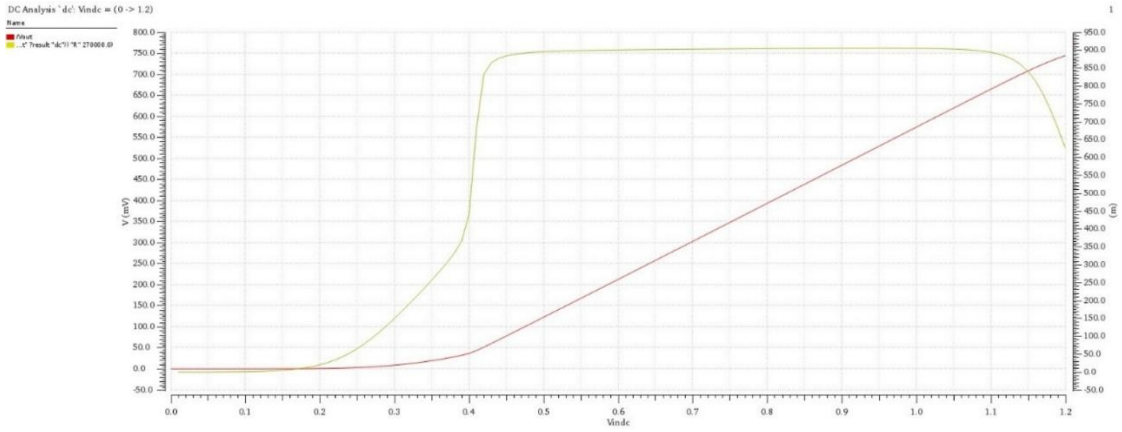


Figure 21: SSF transfer CCs with its derivative (gain) appended on the same plot.

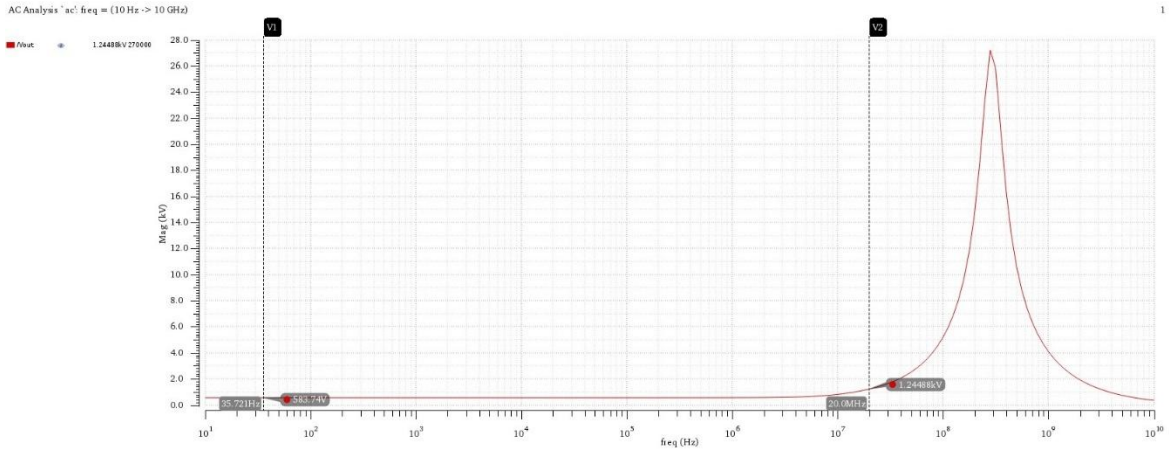


Figure 22: Output impedance frequency response at dc and max in band frequency.

Unfortunately, when connecting each one of these candidates, problems begin to arise. For example, to maintain the feedback device present in the SSF topology (M3) in saturation, its drain-to-gate voltage should be less than the modulus its threshold, while $|V_{GD3}|$ is the same voltage across the channel of the input device (M0), dictating a very small V_{DS0}, which is, clearly, not easy to achieve ($V_{inDC} = V_{outCM} = V_{GS0} + V_{DS1} \approx 0.5\text{V}$. The flipped voltage follower, as another example, suffers from a relatively limited input range (around 200mV), and the resistive feedback inverter amplifier nearly needs a nuclear reactor to get power required to achieve reasonable GBW. The solution for this whole bunch of problems was very trivial: just take the well-designed core amplifier, make a diode connection to implement a balun conversion, make a unity-gain feedback connection to obtain a voltage buffer, and finally optimize the design via running some iterative sweeps. It just worked!

Check Point 2

The following figures illustrate the circuitry used in reporting required measurements in this check point and its CMIR.

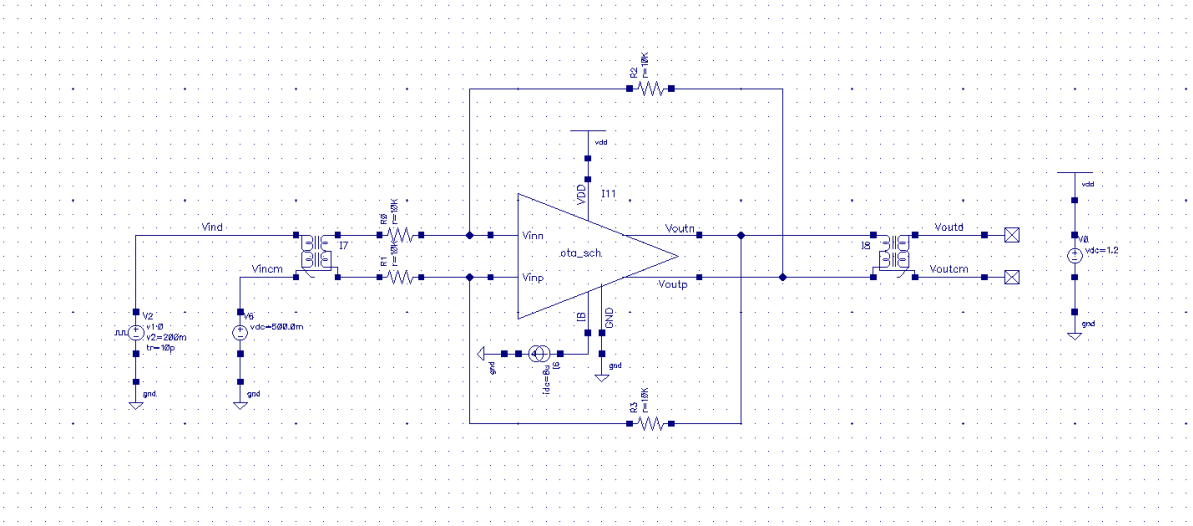


Figure 23: Closed loop testbench.

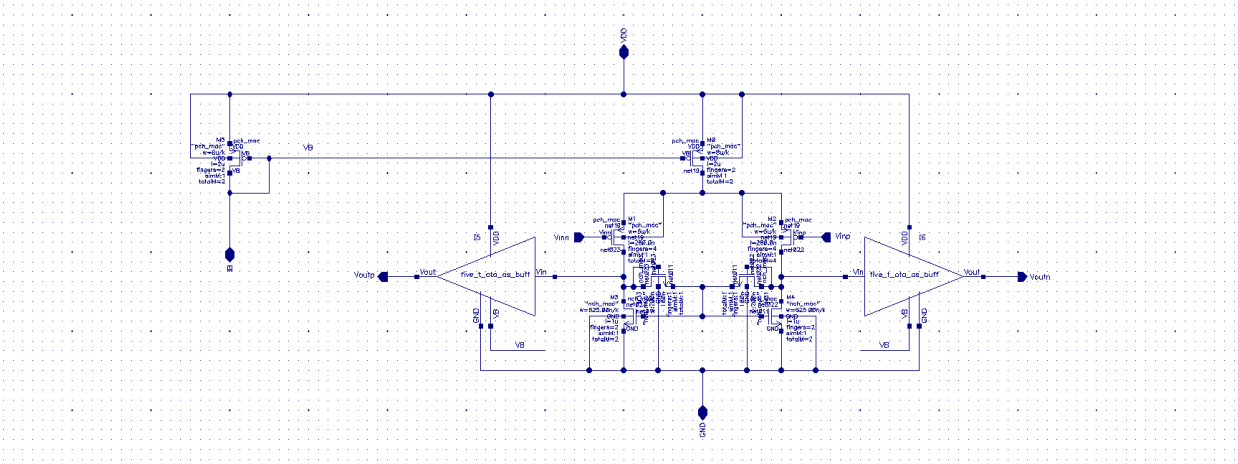


Figure 24: Core Op Amp transistor level.

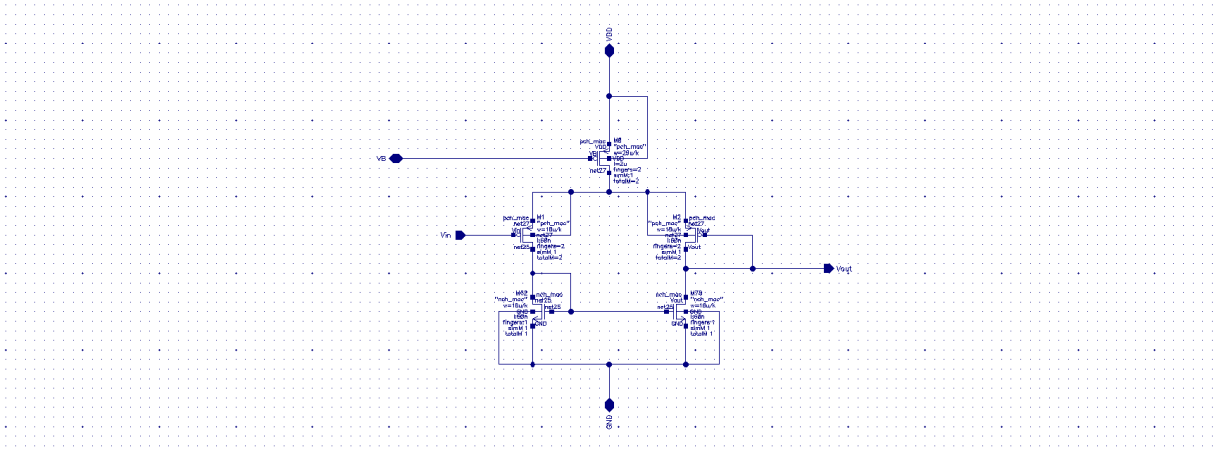


Figure 25: Output stage transistor level.

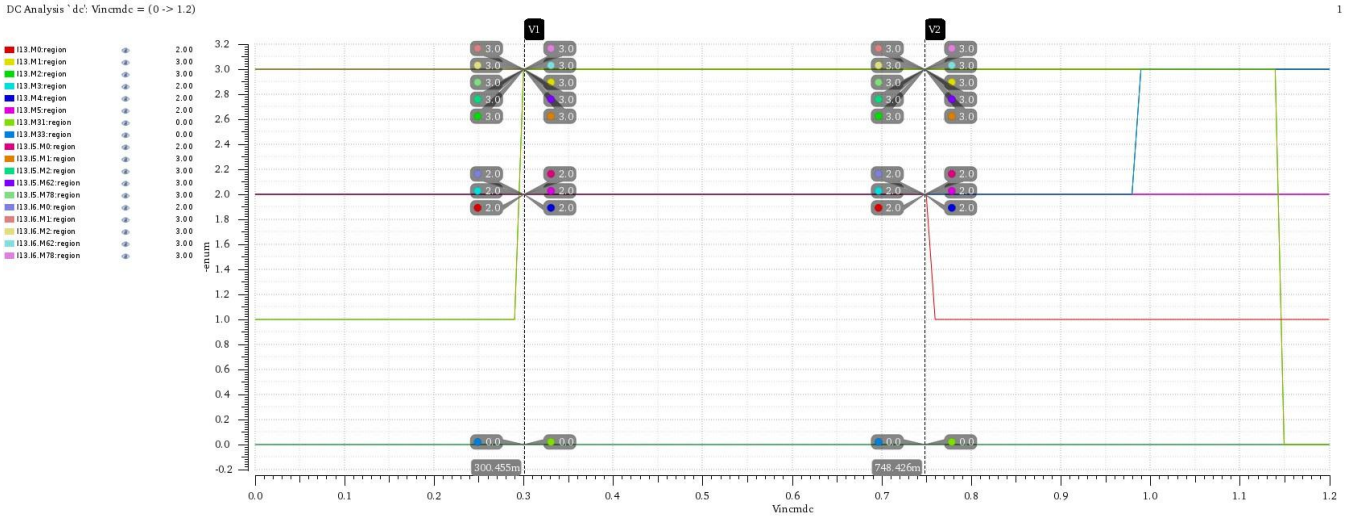


Figure 26: CMIR ≈ 350 mV.



Figure 27: Power consumption.

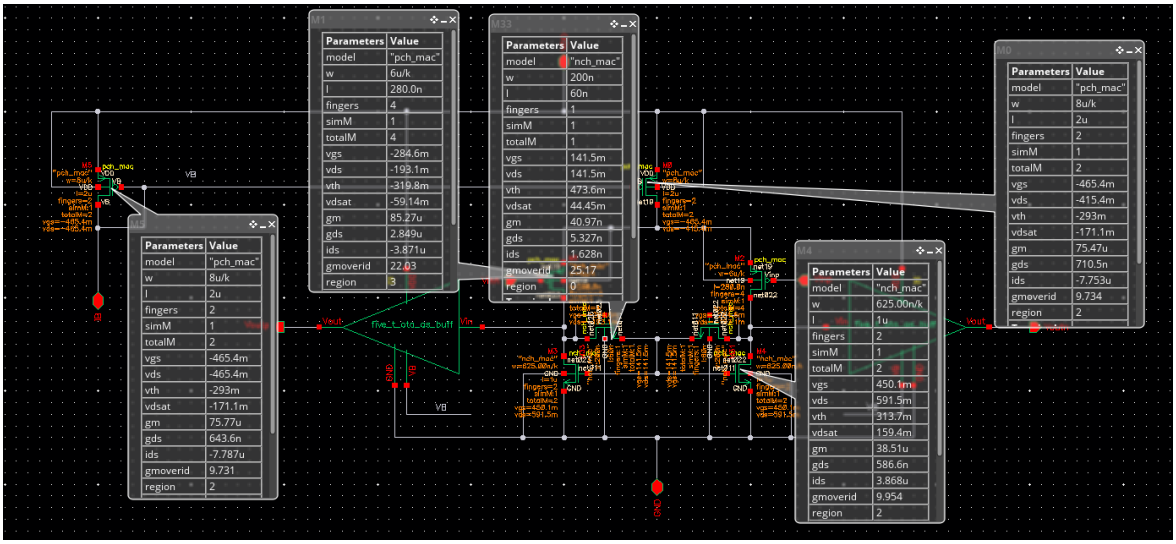


Figure 28: DC OP of the core amplifier

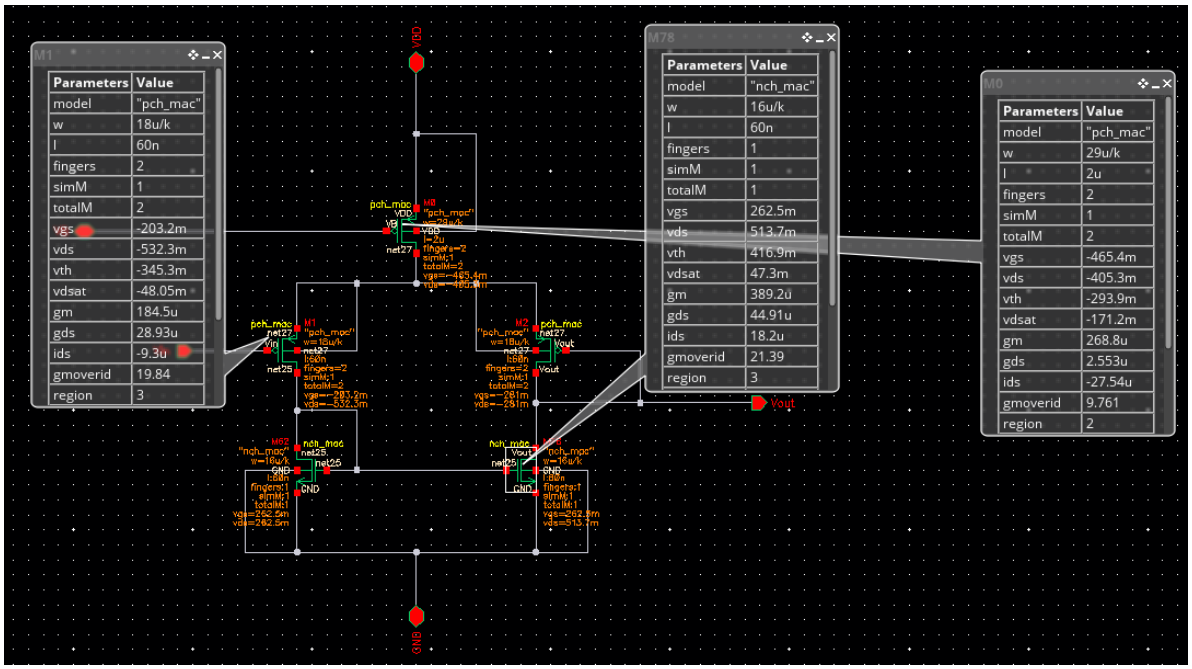


Figure 29: DC OP of the output stage

One of the perspectives that the concept of common mode feedback networks can be viewed from is that they are nothing but a path for any mismatch in current between series-connected current sources. This point of view clarifies how resistors connected in the schematic depicted in figure 16 could successfully realize a CMFB network. However, this CMFB network has one severe limitation: the resistors should be of substantial resistance (10MΩ is used in this work) in order not to degrade the equivalent output resistance and hence the gain, and these large resistors are nearly impossible to layout properly. But, the simplicity of this network that makes everything 10x easier, power saving and lower generated noise (less devices) make it very favorable to use. The solution was also trivial: just drop two transistors of minimum dimensions to minimize parasitic capacitances and diode-connect them to get something that its equivalent electrical model is a resistor as shown in figure 29. It simply also worked as will be shown shortly.

It's required to report the following:

- Differential-mode step response stability in unity-gain feedback.

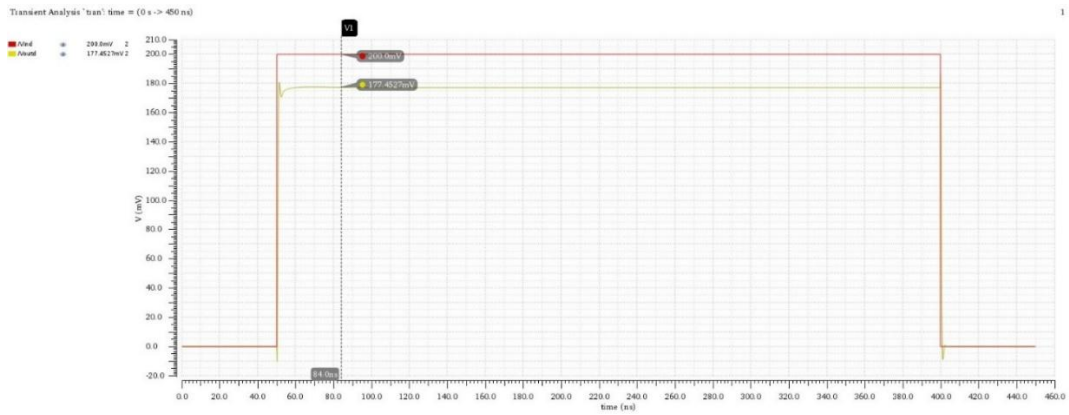


Figure 30: Differential mode step response.

- Common-mode step response stability in unity-gain feedback.

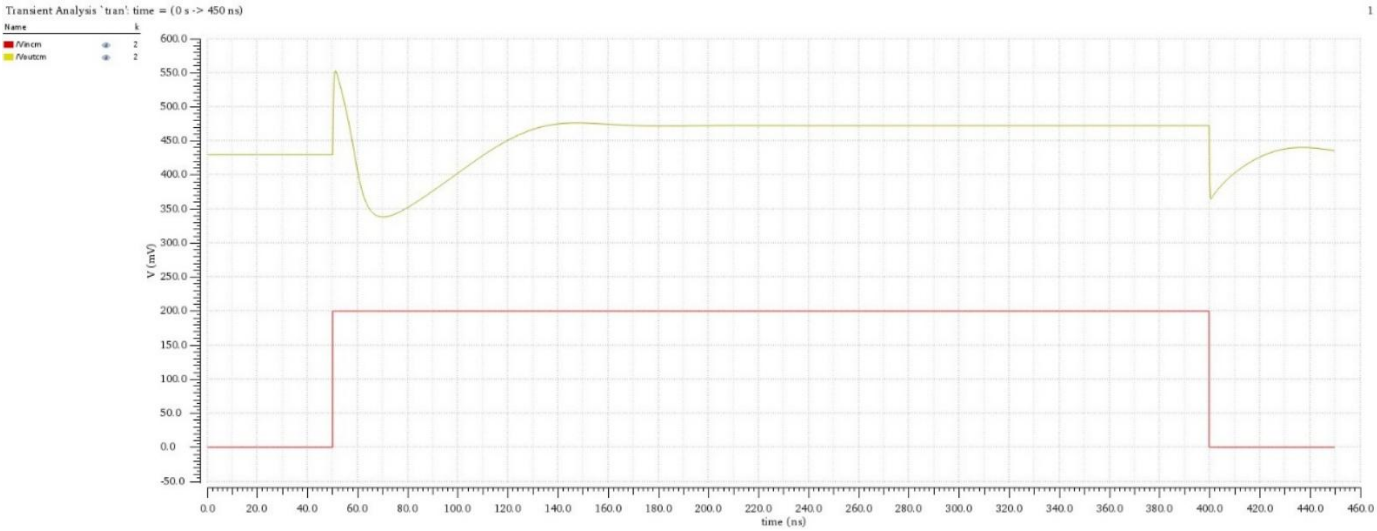


Figure 31: Common mode step response.

- Phase margin larger than 60 degrees in unity-gain.

Test	Output	Nominal	Spec	Weight	Pass/Fail
si_clash_project:check_pt_2:1	PM	66.54			

Figure 32: PM spec overachieved.

- Input noise and output noise spectrum.

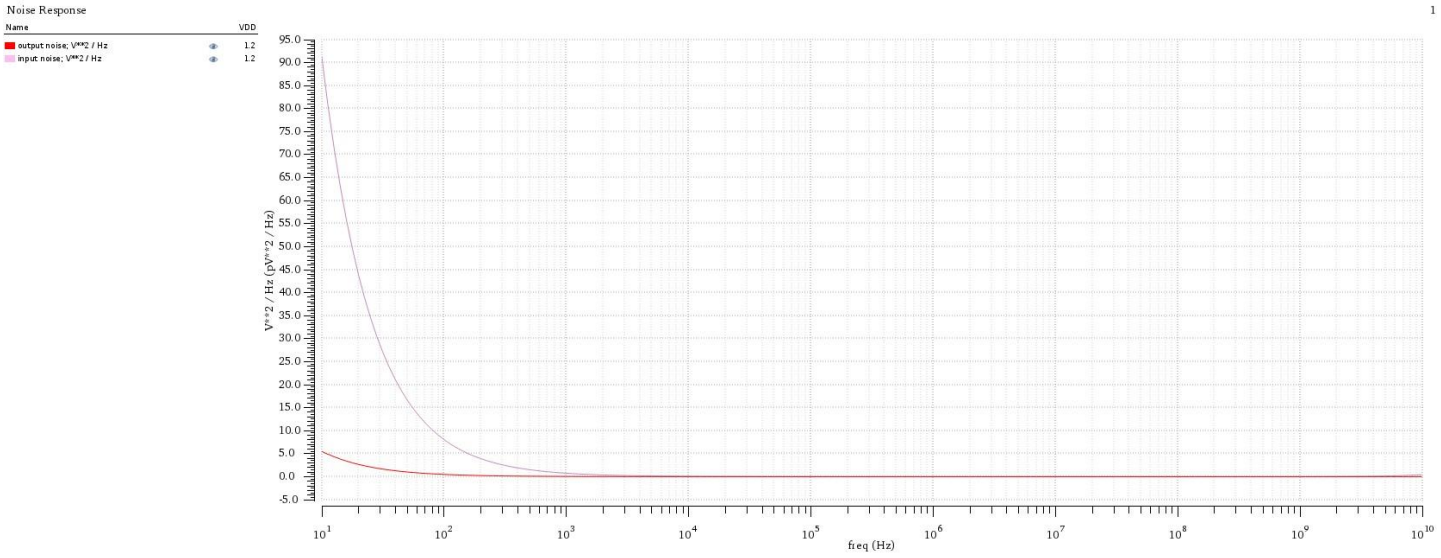


Figure 33: Input & output noise spectrum.

- Comparison of the filter response at the transistor level to the ideal behavioral model, and also to the filter using an ideal op-amp.
    - refer to figure 2, 6, 7 and 36 to observe the schematics used in this deliverable. Also note that the schematic shown in figure 2 is modified to be differential instead of SE to extract accurate noise analysis results).

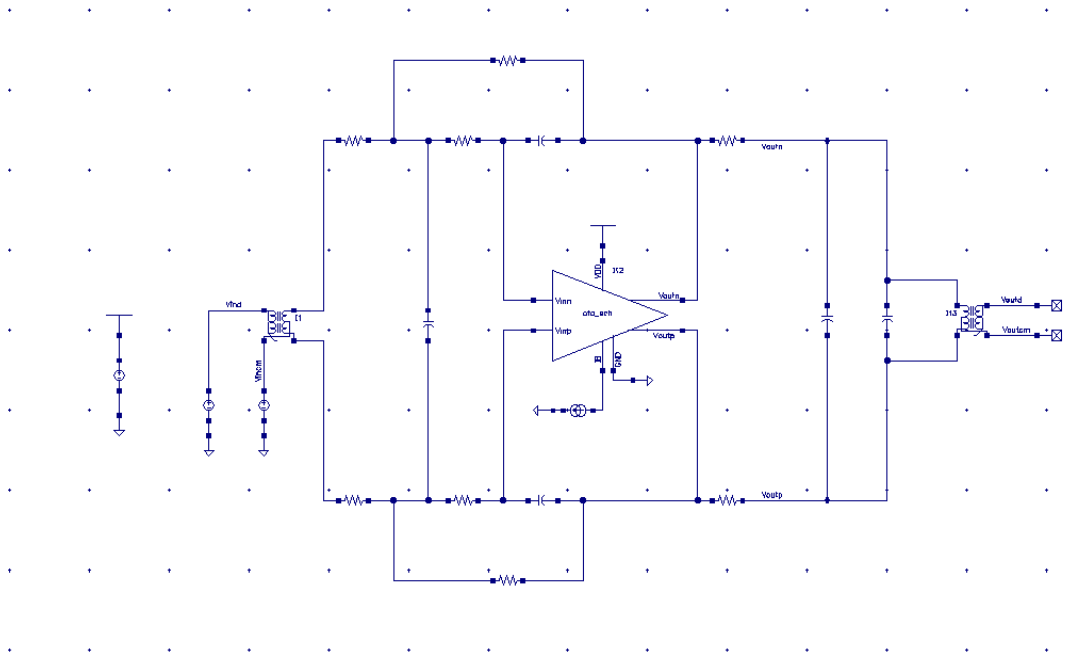


Figure 34: Final circuit design closed loop testbench.

- Frequency response.

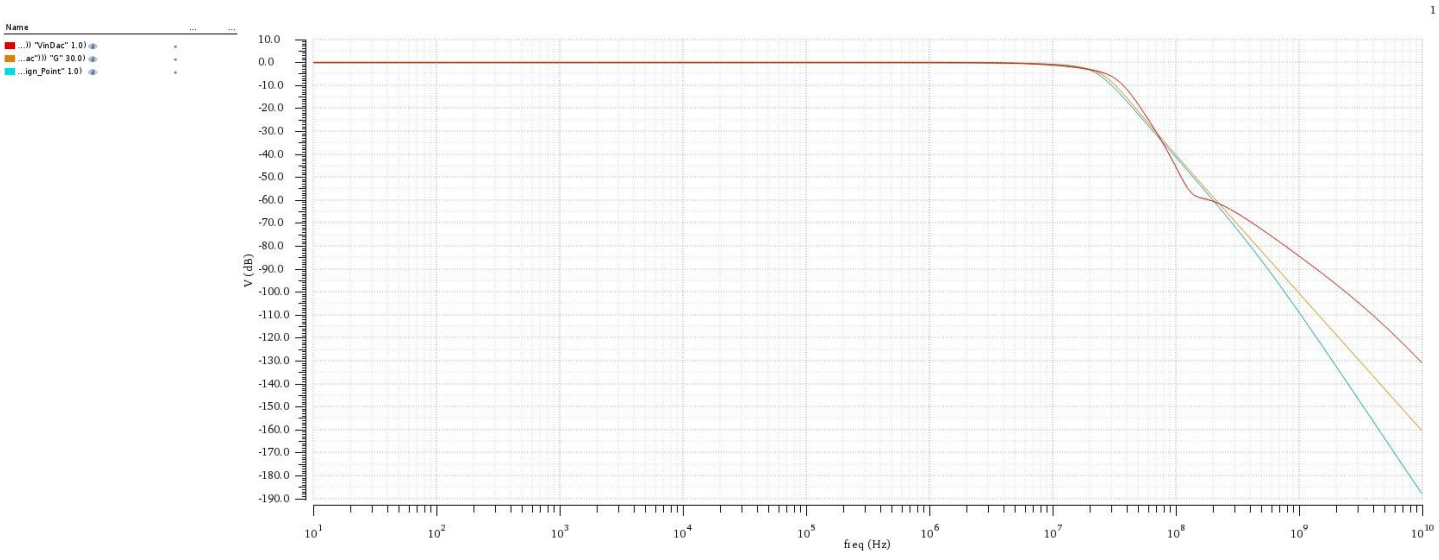


Figure 35: Frequency response of the actual circuit (red), ideal behavioral model (Turquoise) and ideal Op Amp (Orange).

- Step response (200mV differential and common-mode step).

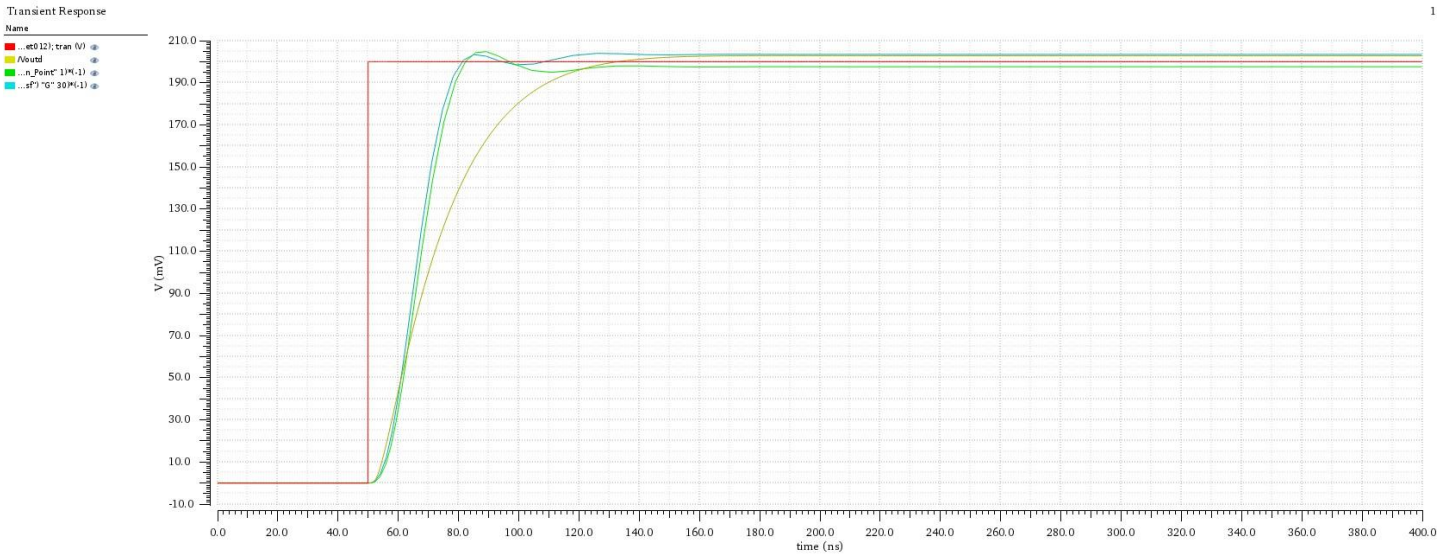


Figure 36: Differential mode step response of actual circuit (Yellow), ideal behavioral model (Green) and ideal Op Amp (Turquoise).

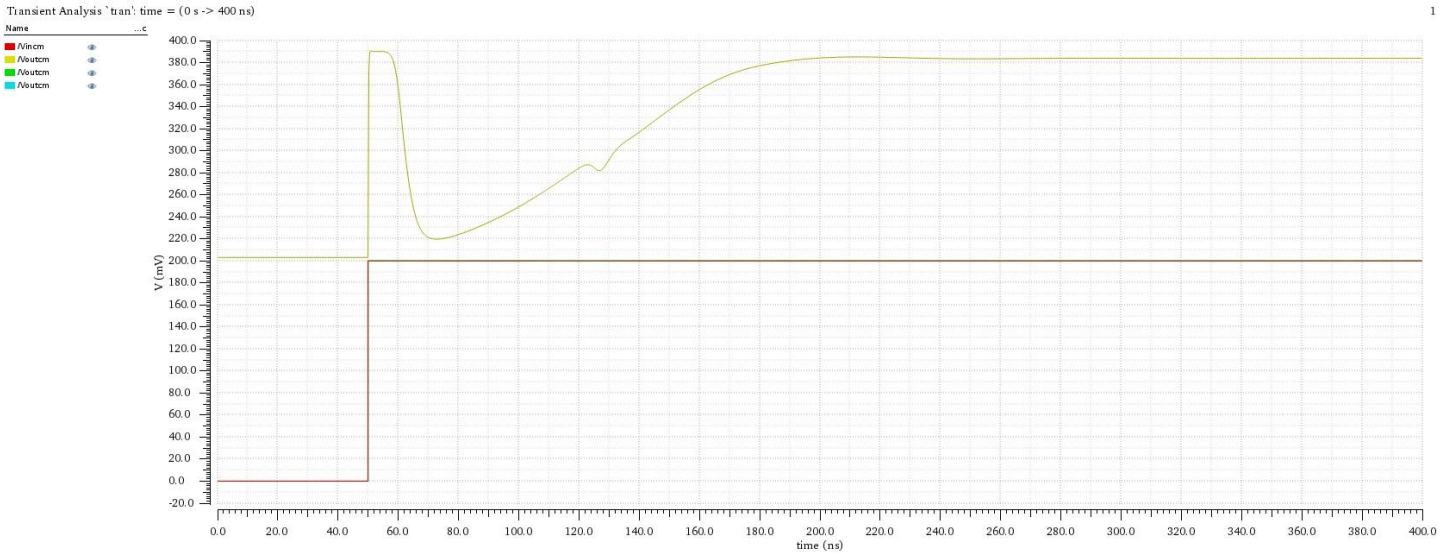


Figure 37: Common mode step response of actual circuit (Yellow), ideal behavioral model (Green) and ideal Op Amp (Turquoise).

- Noise

	VinDac	rmsNoi... 20M)	Design_Point	rmsNoi... 20M)	G	rmsNoi... 20M)
1	1.000	230.7E-6	1.000	133.7E-6	30.00	139.1E-6

Figure 38: Total integrated noise of actual circuit (left), ideal behavioral model (middle) and the ideal Op Amp (right).

- Linearity (THD for 200mV peak value input sinusoid).

	VinDac	thd(v("...2 10M)	Design_Point	thd(v("...2 10M)	G	thd(v("...2 10M)
1	1.000	8.165	1.000	15.64E-3	30.00	16.84E-3

Figure 39: Total harmonic distortion (THD) of actual circuit (left), ideal behavioral model (middle) and the ideal Op Amp (right).

Check Point 3

In this check point, it's required to investigate key performance parameters of the filter when the supply voltage varies by ±10%. Supply voltage is swept from 1.08V (90% of supply) to 1.32V (110% of supply) with 5 mV step.

BW

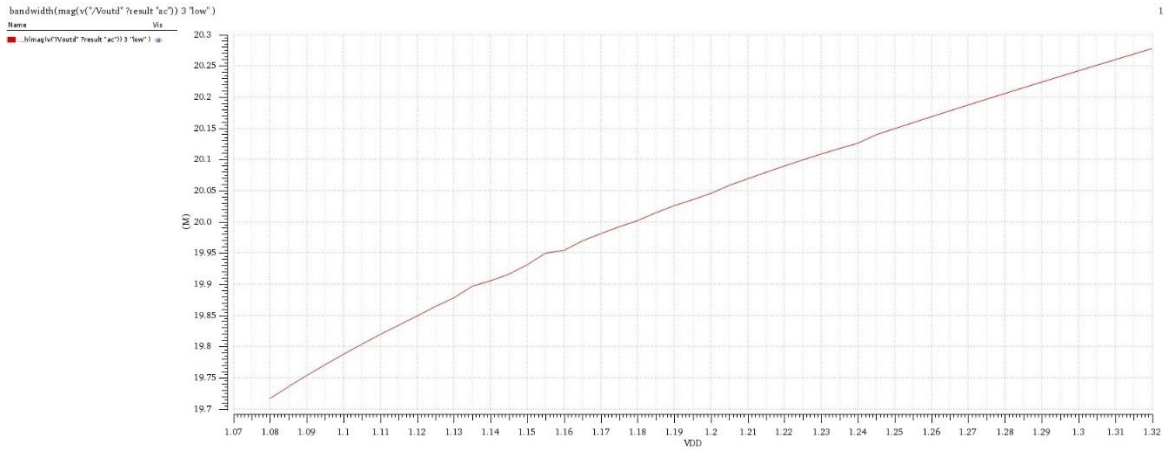


Figure 40: BW variation Vs supply variation

	Expression	Value
1	ymax(bandwid...	560.0E3

Figure 41: ymax – ymin of the previous curve. Fractional variation $=\frac{560k}{20M}\% \approx 2.8\%$

Out-of-Band Attenuation @200MHz

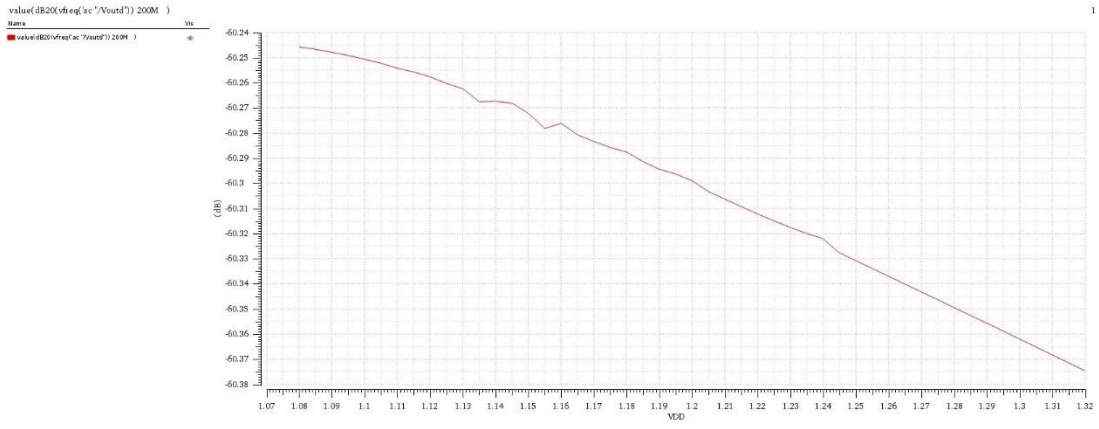


Figure 42: Out-of-band attenuation @200MHz variation Vs supply variation.

	Expression	Value
1	ymax(value(dB...	128.8E-3

Figure 43: ymax – ymin of the previous curve. Fractional variation $=\frac{130m}{55} \times 10^6 \approx 2360\text{ppm}$

Max In Band Group Delay

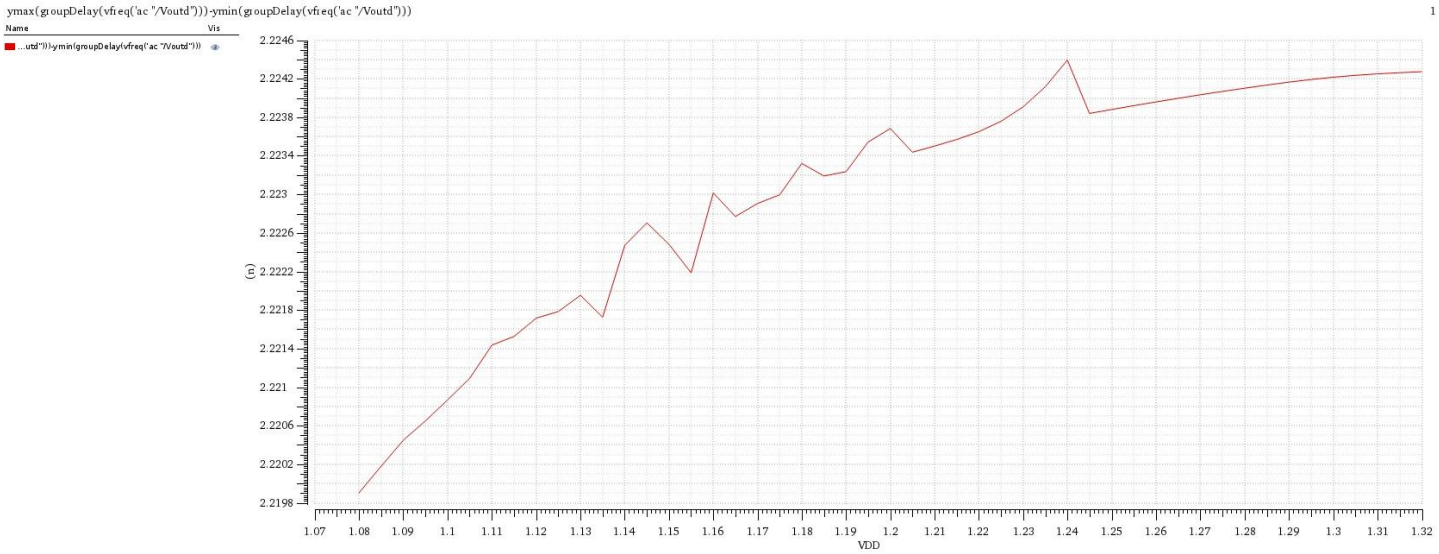


Figure 44: Max in band group delay variation Vs supply variation

	Expression	Value
1	ymax(ymax(gr...	4.487E-12

Figure 45: ymax – ymin of the previous curve. Fractional variation $=\frac{4.5p}{3n}\% \approx 0.15\%$

Dynamic Range

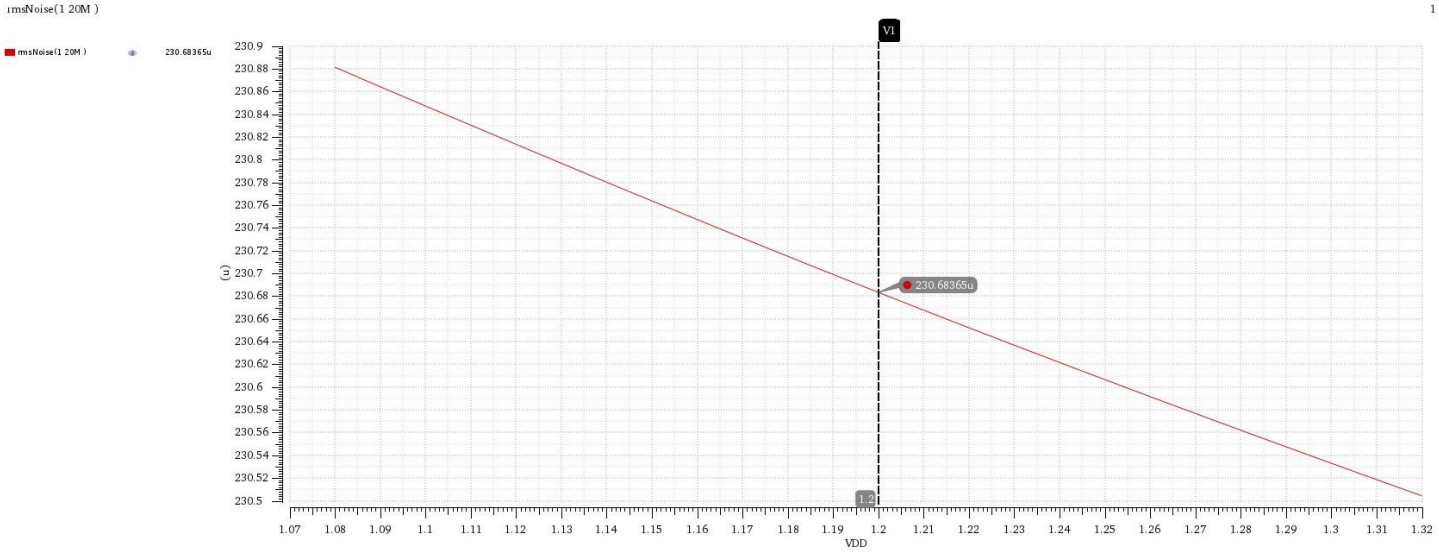


Figure 46: RMS noise variation Vs supply voltage variation

Since supply voltage variation has almost nothing to do with RMS noise, as depicted in the previous plot, the RMS noise at the nominal supply value is assumed in dynamic range calculation, which is, in turn, is defined as the RMS value of the input signal (which is assumed to be a sinusoid 200mV peak-to-peak value) divided by the RMS noise. Then:

$$DR = 20\log\left(\frac{V_{\mathrm{sig}_{\mathrm{rms}}}}{V_{\mathrm{n}_{\mathrm{rms}}}}\right) = 20\log\left(\frac{\frac{100m}{\sqrt{2}}}{230\mu}\right) \approx 49.755\,\mathrm{dB}$$

CMRR

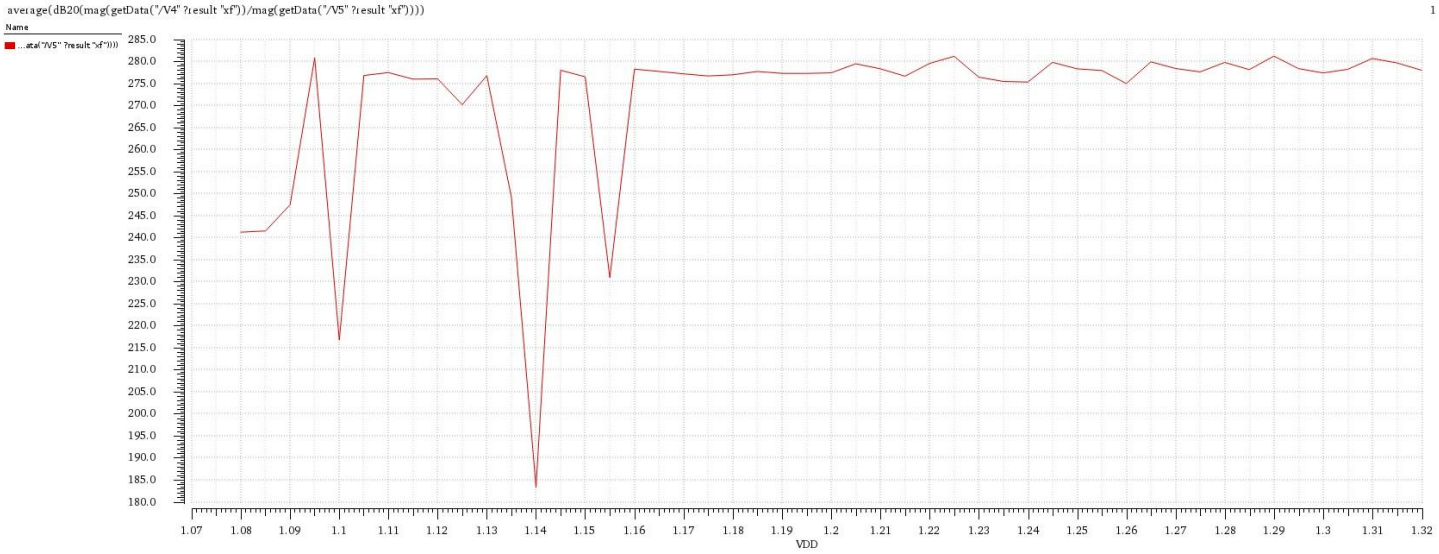


Figure 47: CMRR variation Vs supply voltage variation

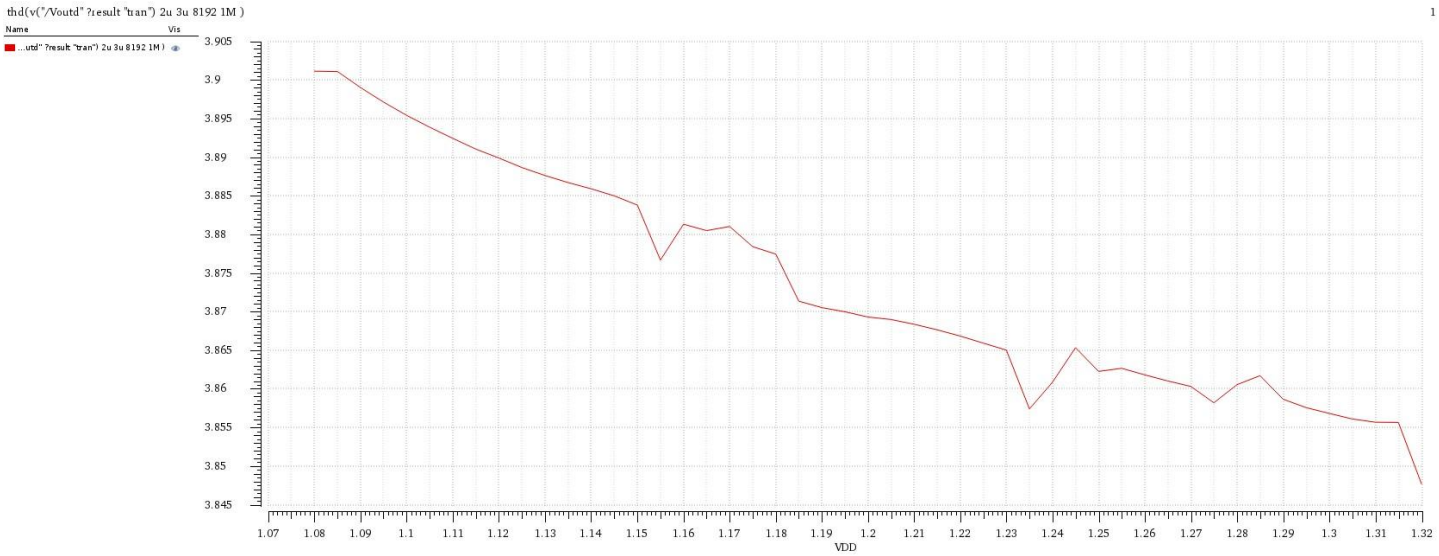


Figure 48: THD % variation Vs supply voltage variation.

The common-mode feedback should be stable (phase margin of 60 degrees). Check stability over the supply voltage variations.

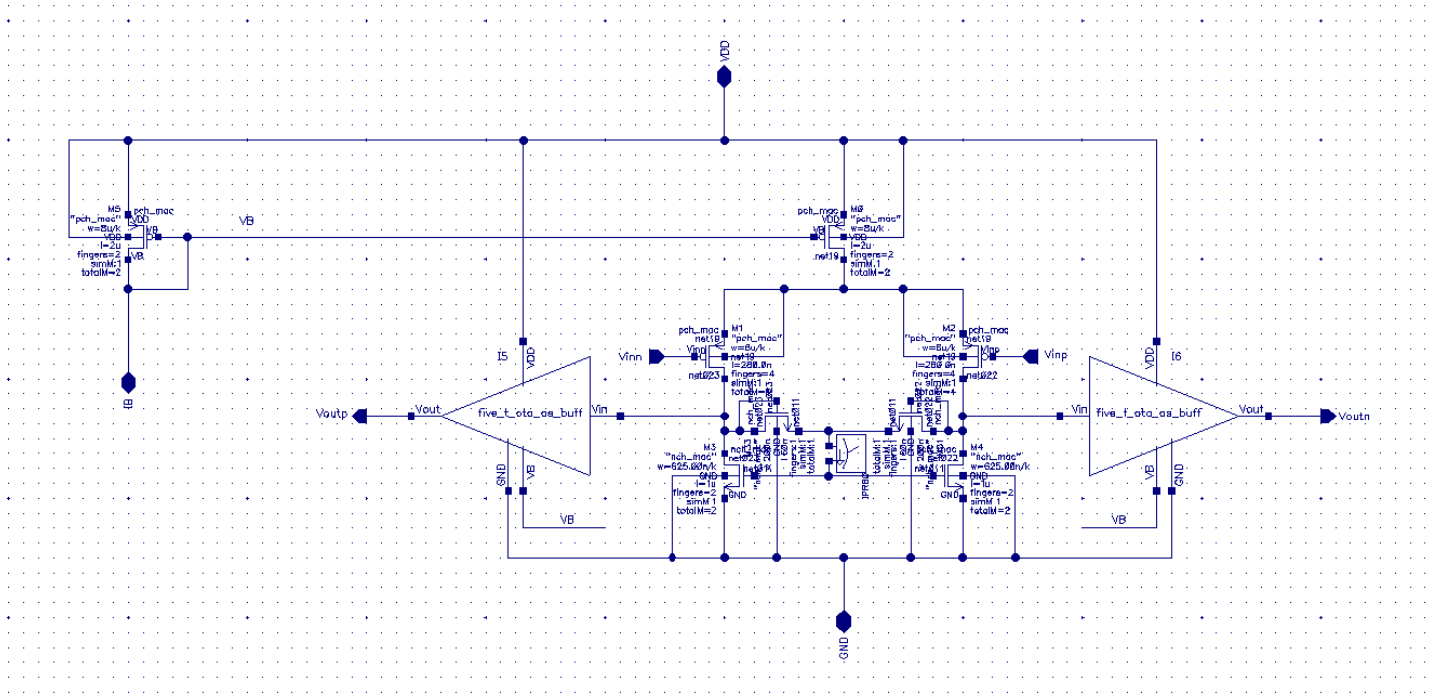


Figure 49: Testbench for stb analysis with iprobe inserted to cut the CM loop.

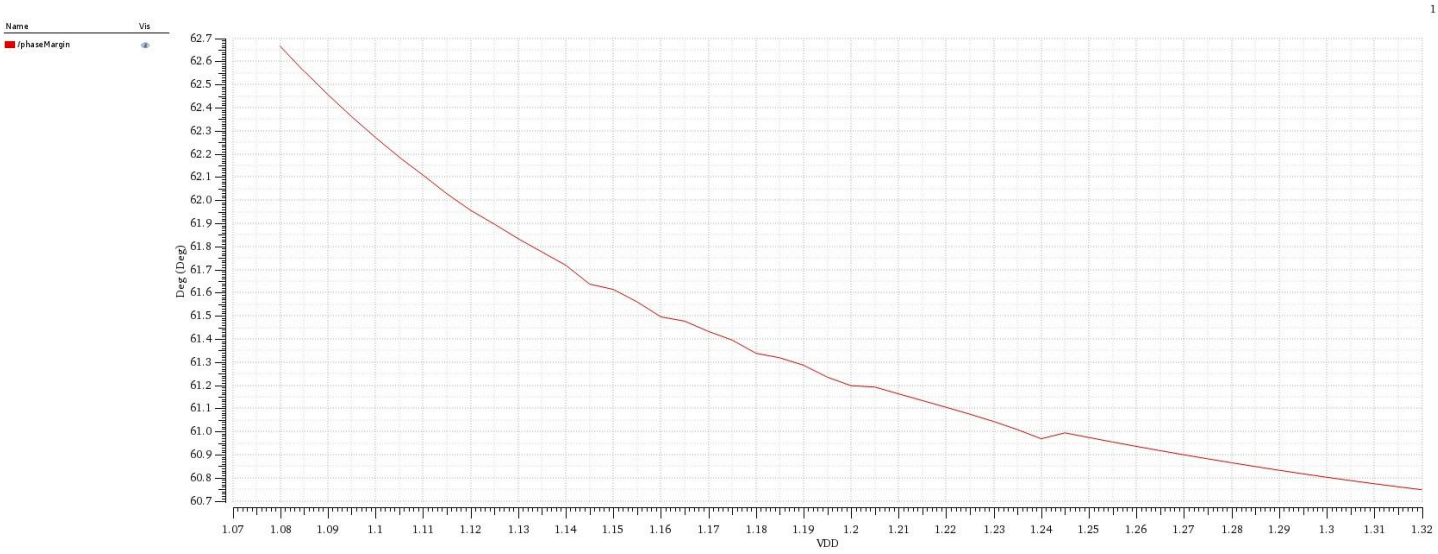


Figure 50: PM variation Vs supply voltage variation

The common-mode rejection of a common mode 400mV (peak-to-peak) signal (transient simulation) should be demonstrated.

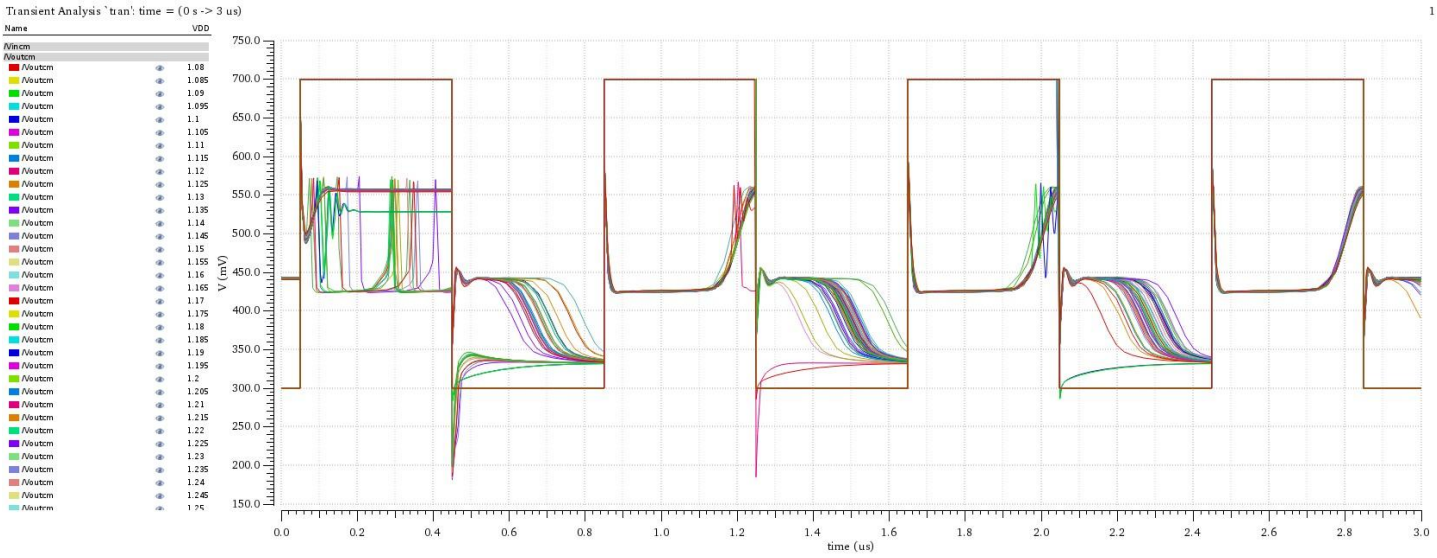


Figure 51: Output CM settling time variation Vs supply voltage variation.

Check Point 5

Following is the VerilogA code for a resistor model that is to be calibrated via another VerilogA code that models a calibration scheme:

```
// VerilogA for tests_65, res_code, veriloga


`include "constants.vams"
`include "disciplines.vams"


module res_code(p,n);
inout p, n;
electrical p,n ;
parameter real Rnom =1e3;
parameter integer code =0;
parameter integer bits =6;
parameter real tune_min =0.5;
parameter real tune_max =1.5;

real mult, y ;
analog begin
y=pow(2,bits);
mult = tune_max+(tune_max-tune_min)*(code/y-1);

V(p,n) <+ I(p,n)*Rnom*(1+mult);
end
endmodule
```

and the code of a capacitor model:

```
// VerilogA for tests_65, tune_cap_code, veriloga


`include "constants.vams"
`include "disciplines.vams"


module tune_cap_code(p,n);


inout p, n;
electrical p,n ;
parameter real Cnom =1e-3;
parameter integer code =0;
parameter integer bits =6;
parameter real tune_min =0.5;
parameter real tune_max =1.5;


real mult ,Cavl, y;
analog begin
y=pow(2,bits);
mult = tune_max+(tune_max-tune_min)*(code/y-1);
Cavl=Cnom*(1+mult);
I(p,n)<+ Cavl*ddt(V(p,n));
end
endmodule
```

and finally, a VerilogA model of the Op Amp:

```
// VerilogA for si_clash_verilog, op_amp_code, veriloga


`include "constants.vams"
`include "disciplines.vams"


module op_amp_code(inp,inn,out);
input inp ,inn;
output out;
electrical inp ,inn ,out ,cout;


parameter real  Ao=1e4 from (0:inf);
parameter real  GBW =10e6 from (0:inf);
parameter real  Rout =50 from [0:inf);
parameter real  Vos=0;
```

```
real tau [0:1] ,p[0:1];
real Vdiff ,Vout_int;

analog begin
tau[1]=Ao/(2*GBW*`M_PI);
tau[0]=1;;
Vdiff=V(inp,inn)+Vos;
V(cout)<+ laplace_nd(Vdiff,{Ao},tau );
// laplace_nd(Ao*Vdiff,{1},tau );
I(cout,out ) <+ V(cout,out)/Rout;
end
endmodule
```

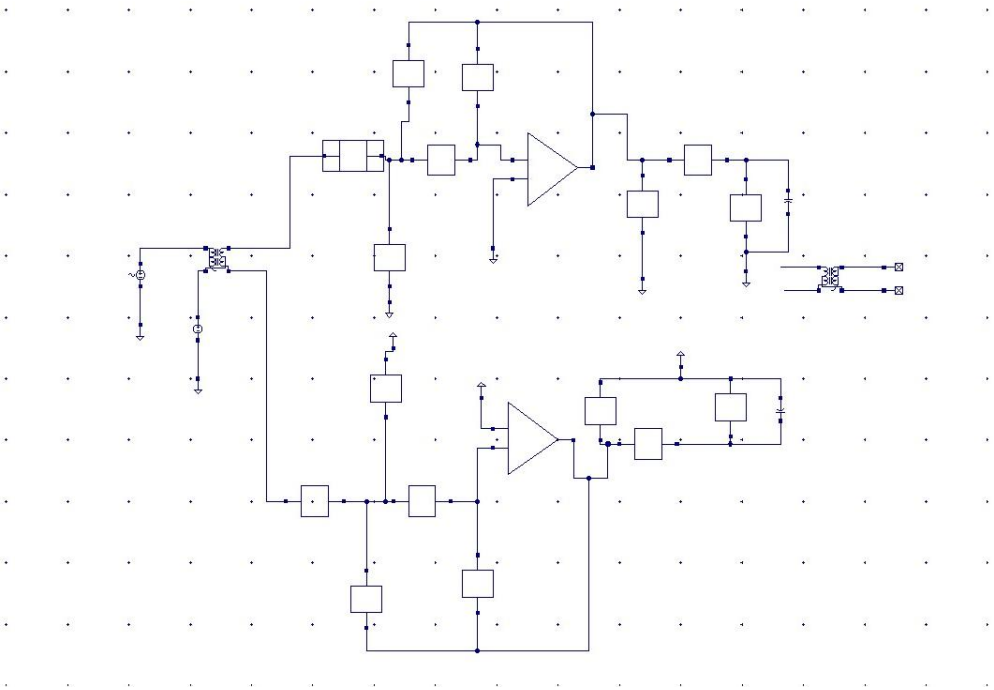


Figure 52: Circuit testbench built using VerilogA models stated above

Test	Output	Nominal	Spec	Weight	Pass/Fail
Si_project:verilog_test_becnch:1	/VOP				
Si_project:verilog_test_becnch:1	/VON				
Si_project:verilog_test_becnch:1	/VOUT				
Si_project:verilog_test_becnch:1	groupDelay(VF("/VOP"))				
Si_project:verilog_test_becnch:1	bandwidth(VF("/VOP") 3 "low")	20.01M			
Si_project:verilog_test_becnch:1	dB20(VF("/VOP"))				
Si_project:verilog_test_becnch:1	groupDelay(VF("/VOUT"))				
Si_project:verilog_test_becnch:1	bandwidth(VF("/VOUT") 3 "lo...	20.01M			
Si_project:verilog_test_becnch:1	gainBwProd(VF("/VOUT"))	17.19M			
Si_project:verilog_test_becnch:1	dB20(VF("/VOUT"))				

Figure 53: Circuit performance without any mismatch

Test	Output	Nominal	Spec	Weight	Pass/Fail
Si_project:verilog_test_becnch:1	/VOP				
Si_project:verilog_test_becnch:1	/VON				
Si_project:verilog_test_becnch:1	/VOUT				
Si_project:verilog_test_becnch:1	groupDelay(VF("/VOP"))				
Si_project:verilog_test_becnch:1	bandwidth(VF("/VOP") 3 "low")	20.27M			
Si_project:verilog_test_becnch:1	dB20(VF("/VOP"))				
Si_project:verilog_test_becnch:1	groupDelay(VF("/VOUT"))				
Si_project:verilog_test_becnch:1	bandwidth(VF("/VOUT") 3 "lo...	23.12M			
Si_project:verilog_test_becnch:1	gainBwProd(VF("/VOUT"))	15.91M			
Si_project:verilog_test_becnch:1	dB20(VF("/VOUT"))				

Figure 54: Circuit performance with some mismatch introduced

- Other key performance parameters do not change that much in accordance to introduced mismatch.

Check Point 6

All work and effort exerted in check point 6 can be found via this link

Conclusions

Following is a table containing a summary of required and achieved specs.

Spec	**Required**	**Achieved**
Gain & BW	0dB & 20MHz	0dB & 20MHz :)
Out-of-band attenuation @200MHz	55dBc	≈60dB :)
Max in band group delay	3nsec	≈2.2nsec :)
Dynamic range	50dB	49.755dB
THD	1%	3.9% :(
Power consumption	Minimum	85μW :)

Performance Metrics Across Temperature Variation

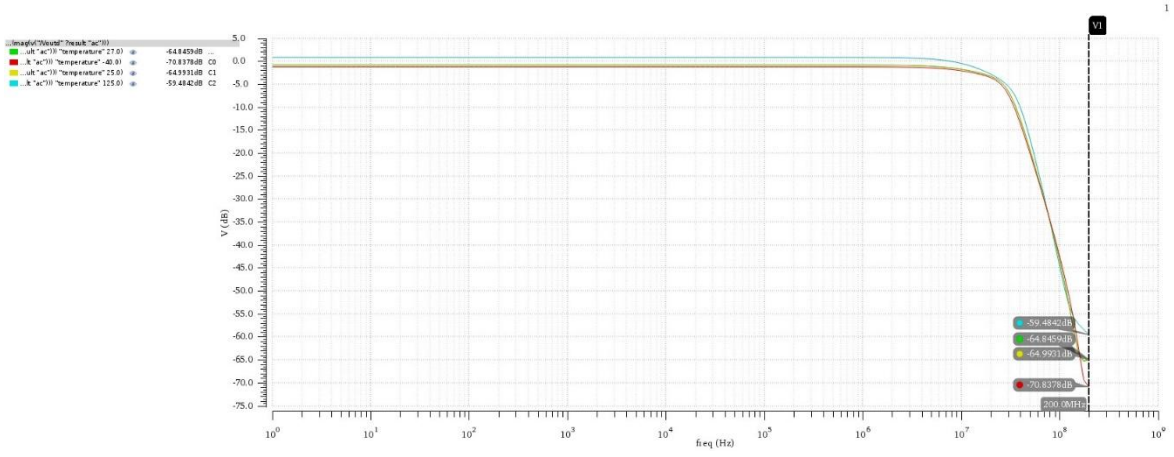


Figure 55: Frequency response across temperature corners

	temperature	ymax(gr...utd")))
1	-40.00	1.406E-9
2	25.00	1.692E-9
3	27.00	1.704E-9
4	125.0	2.541E-9

Figure 56: Max in band group delay across temperature corners

	temperature	thd(v("...92 1M)
1	-40.00	3.707
2	25.00	6.732
3	27.00	7.154
4	125.0	31.94

Figure 57: THD across temperature corners

	temperature	rmsNoi... 20M)
1	-40.00	172.3E-6
2	25.00	203.2E-6
3	27.00	204.2E-6
4	125.0	265.0E-6

Figure 58: RMS noise of CL Op Amp across temperature corners

References

1. Chapter 8, Analog Filter Design, Analog Devices.
2. Antaryami Panigrahi, Abhipsa Parhi: '1.2 V, 12.5 MHz fourth-order low-pass filter with 83 dB stopband attenuation using low output impedance source follower in 45 nm CMOS'.
3. Dr. Hesham Omran, CMOS Analog IC Design, ITI.
4. Dr. Hesham Omran, Analog Integrated Systems Design, Filters lecture.
5. Tons of other Reddit and electronics forum threads and some random papers :).